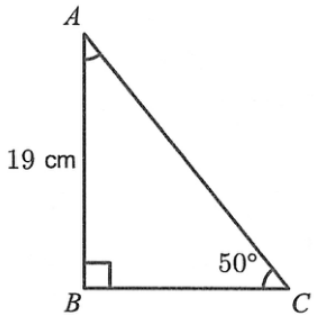
Trigonometry: Introduction

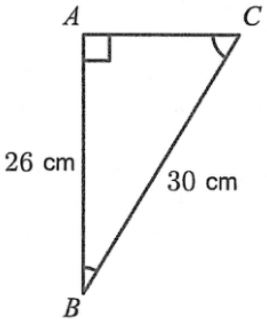
Questions 1–21

Question 1. Find all the missing angles and sides in the following right-angled triangles, giving your answers to three significant figures.

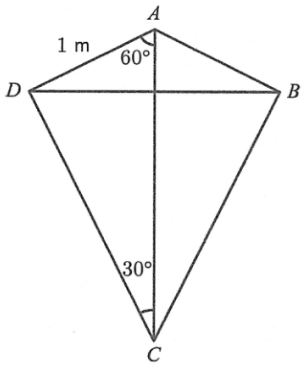
a)



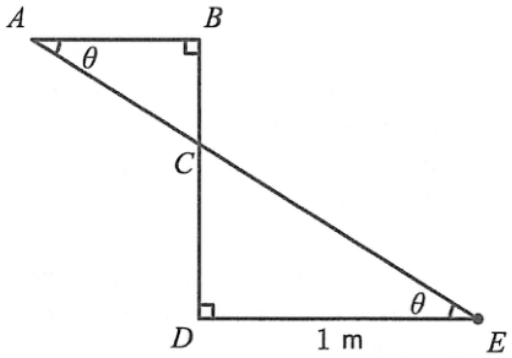
b)



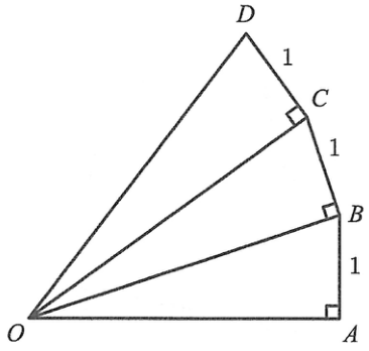
Question 2. The diagram below shows the kite $ABCD$, where angle $DAC = 60°$ and angle $ACD = 30°$. It is also given that the length of AD is 1 m. Find the exact area of the kite.



Question 3. In the diagram below, the angles ABC and CDE are both 90°, and the length DE is 1 m. It is given that BCD is a straight line. The area of triangle ABC is one third of the area of triangle CFE. Find the exact length of AB.

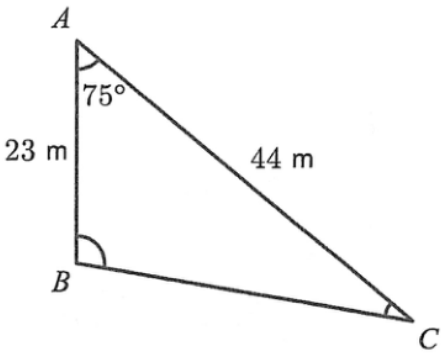


Question 4. In the diagram below, the angles OAB, OBC and OCD are all 90°, and the lengths AB, BC and CD are all 1 unit.



Find the length OA given that $\frac{OD}{OA} = \sqrt{2}$.

Question 5. Find the area of the triangle shown in the diagram below, giving your answer to three significant figures.



Question 6. It is given that ABC is a triangle where angle ABC is $45°$, side $AB = 20$ cm, and side $BC = 15$ cm. Find the length of side AC.

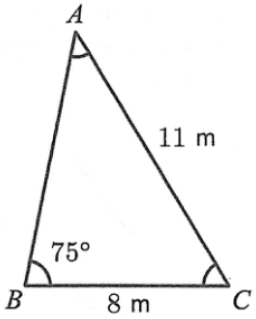
Question 7. It is given that ABC is a triangle where AB is 9 cm, BC is 12 cm, and angle $ACB = 40°$.

a) Use the sine rule to find two possible values for angle BAC, giving your answers to one decimal place.

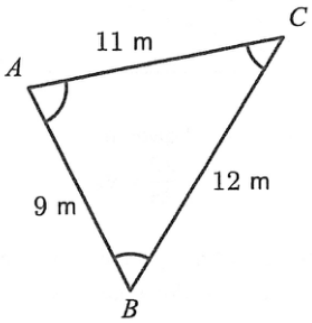
b) Hence draw and label two possible triangles ABC.

Question 8. Find all the missing angles and sides in the following triangles, giving your answers to three significant figures.

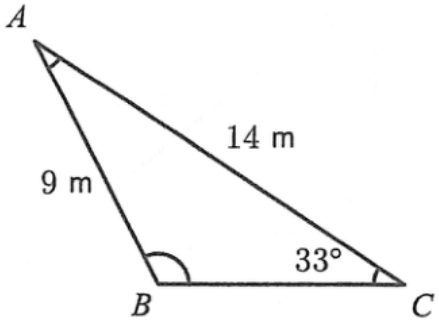
a)



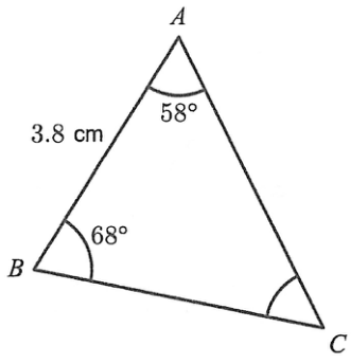
b)



c)

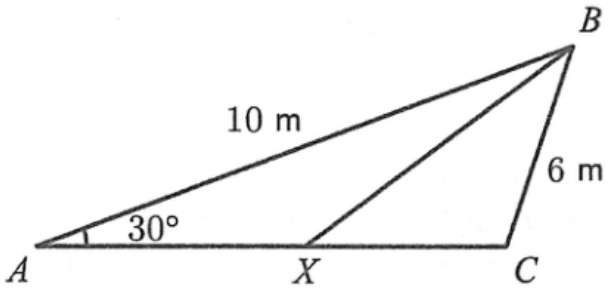


Question 9. Find the area of the triangle shown in the diagram below, giving your answer to three significant figures.



Question 10. It is given that ABC is a triangle where AB is 16 cm, AC is 9 cm and angle BAC is $x°$. The area of the triangle is 25 cm^2. Find the two possible values of x to one decimal place.

Question 11. The diagram below shows the triangle ABC. It is given that angle ACB is obtuse. Find the length of XC given that BX bisects the triangle.



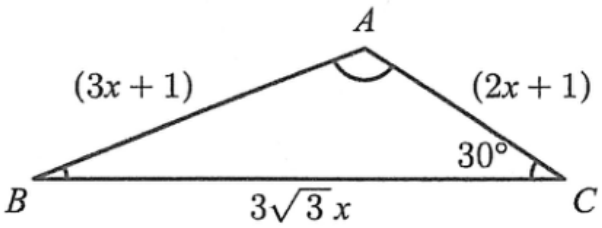
Question 12. It is given that ABC is a triangle where AB is $5m$ cm, AC is $2m^2$ cm and angle BAC is $45°$. The area of the triangle is 640 cm^2. Show that $m = k\sqrt{2}$ where k is an integer to be found.

Question 13. The diagram below shows the triangle ABC.

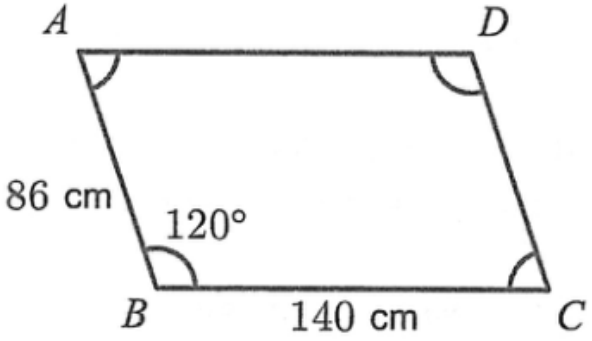
a) Show that $11x - 4x^2 = 0$

b) Find the value of x

c) Find the largest angle in the triangle.

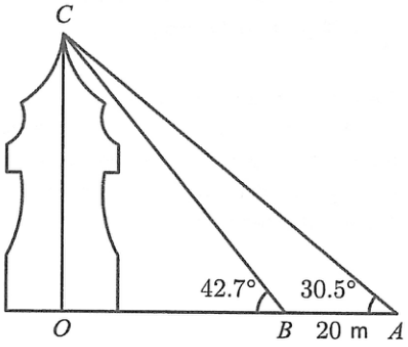


Question 14. The diagram below shows a parallelogram with sides of length 86 cm and 140 cm. It is given that angle $ABC = 120°$. Show that the exact area of the parallelogram is $k\sqrt{3}$ m^2 where k is a rational number.



Question 15. It is given that a triangle has sides of length b, $b+1$ and $b+2$, where $b > 0$. The largest angle in the triangle is labelled x. Use the cosine rule to show that $\cos x = \frac{b-3}{2b}$.

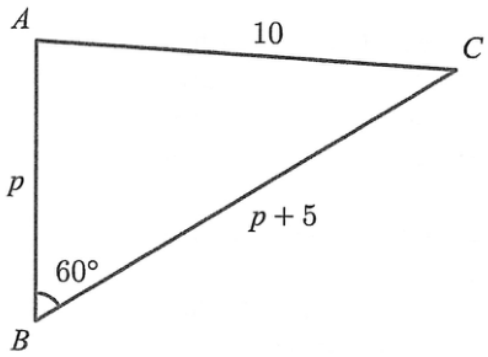
Question 16. Ben measures the angle of elevation from where he is standing to the top of a tall building to be 30.5°. He then walks straight towards the building for 20 metres, stops, and measures the angle of elevation again. This time it is 42.7°. Determine the height of the building.



Question 17. Three archaeological digs, A, B and C, are drawn on a map. B is 180 metres due South of A. C is 320 metres from A on a bearing of 075°. Find

a) the bearing of B from A
b) the bearing of A from C
c) the bearing of B from C

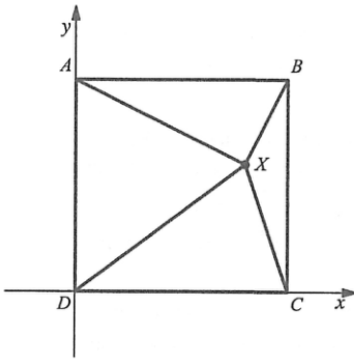
Question 18. The diagram below shows a triangle, ABC, with sides of length p, $p+5$ and 10. $\angle ABC = 60°$. Find the exact value of p.



Question 19. The area of a regular pentagon is 60 cm^2. Calculate the length of one of its sides.

Question 20. The area of a regular octagon is 450 cm^2. Calculate the length of one of its sides.

Question 21. The diagram below shows a square, $ABCD$, with area 100 units^2 drawn on the coordinate axes. The square has been divided up into four triangles, XBC, XAB, XCD, and XAD, with areas 10 units^2, 20 units^2, 30 units^2, and 40 units^2 respectively.



a) X has exact coordinates (p, q). Find the value of p and the value of q
b) Y has exact coordinates (r, q), where $0 < r < p$. The triangle XYD has an area that is at least half of the area of triangle XAD. Find the range of possible values of r